MDM

Subject :- ME24052 – Introduction to Robotics & Automation

Topic:- Fundamentals of Robot Programming and Applications



Mr. Shrikant C. Mahadik
Assistant Professor
Department of Mechanical Engineering ,
VPKBIET Baramati
Mob No:- 9850090011
Email- shrikant.mahadik@vpkbiet.org

Unit-4: Fundamentals of Robot Programming and Applications	**[08 Hrs.]**
Introduction to Robotic Programming, On-line and off-line programming, programming examples. Various Teaching Methods, Survey of Robot Level Programming Languages with manufacture, Various Textual Robot Languages, Typical Programming Examples such as Palletizing, Job Loading & unloading, Welding, Press Work Etc. Robots in manufacturing and non-manufacturing applications, a robot-based manufacturing system, Robot controllers types & specification.	

Robotic programming

- Robotic programming is the process of creating a sequence of instructions for a robot to perform specific tasks.
- These programs can be written using specialized robot programming languages or through a combination of textual programming and teaching methods.
- Robotic programming enables robots to perform complex operations in manufacturing, assembly, and other industries.

On-Line and Off-Line Programming

1. On-Line Programming

- Involves directly controlling and teaching the robot using a **teach pendant** or joystick while it is in operation.
- The robot learns movements and stores them as programs for future execution.

Advantages:

- Real-time feedback allows for immediate corrections.
- Simple to implement for repetitive tasks.

Disadvantages:

- The robot cannot be used for production during programming.
- Time-consuming for complex tasks

2. Off-Line Programming

- Programs are created in a virtual environment, often using simulation software, and then uploaded to the robot.
- Allows for testing and optimizing the robot's movements without halting production.

Advantages:

- Minimizes downtime.
- Enables complex simulations and optimization.
- Reduces programming errors.

Disadvantages:

- Requires advanced software and expertise.
- Accuracy depends on the quality of the simulation.

Various Teaching Methods

- **Manual Teaching**:
  - Using a teach pendant or guiding the robot arm manually to record positions.
- **Lead-Through Teaching**:
  - The operator physically moves the robot to the desired positions, and the robot records these movements.
- **Demonstration-Based Teaching**:
  - The robot observes human actions and replicates them using AI-based algorithms.
- **Sensor-Based Teaching**:
  - The robot uses sensory feedback (e.g., force or vision) to learn the task.
- **Offline Teaching**:
  - Programming in a virtual environment without direct interaction with the robot.

Survey of Robot Level Programming Languages with Manufacturers

- **Val II (Unimation)**:
  - Early robotic language used in Unimation robots for simple task automation.
- **Karel (Fanuc)**:
  - Proprietary language by Fanuc, used for programming advanced robot tasks.
- **RAPID (ABB)**:
  - Advanced language by ABB, enabling high-level programming of industrial robots.
- **V+ (Adept Technology)**:
  - Used for precise and flexible robot control in Adept robots.
- **KUKA Robot Language (KRL)**:
  - Language for programming KUKA robots, emphasizing modularity and flexibility.
- **MELFA BASIC (Mitsubishi)**:
  - Used for programming Mitsubishi industrial robots, with a focus on ease of use.
- **URScript (Universal Robots)**:
  - Lightweight and user-friendly, designed for collaborative robots.

Various Textual Robot Languages

- **Assembly-like Languages**:
  - Early languages with limited abstraction, requiring detailed motion commands.
- **Structured Text Languages**:
  - High-level languages resembling general-purpose programming languages (e.g., Python-like syntax).
- **Graphical Programming**:
  - Visual interfaces for defining robot tasks without textual coding.
- **Domain-Specific Languages**:
  - Tailored to specific robot applications, such as palletizing or welding.

Robots in Manufacturing and Non-Manufacturing Applications

1. Manufacturing Applications

- **Material Handling**: Loading, unloading, and transferring parts.
- **Assembly**: High-speed assembly of electronic components or automotive parts.
- **Welding**: Precision spot or arc welding in automotive industries.
- **Painting and Coating**: Ensures uniform coverage and minimizes waste.
- **Quality Inspection**: Vision-guided robots for defect detection.

Non-Manufacturing Applications

- **Healthcare**: Surgical robots and rehabilitation devices.
- **Agriculture**: Robots for planting, harvesting, and sorting.
- **Logistics**: Warehouse robots for picking and packing.
- **Space Exploration**: Robots for planetary exploration and satellite servicing.
- **Defense**: Bomb disposal and reconnaissance robots.

Robot-Based Manufacturing Systems

- **Description**: Systems where robots handle tasks like assembly, welding, or packaging in a production line.
- **Advantages**:
  - Increases efficiency and consistency.
  - Reduces human error.
  - Enhances safety in hazardous environments.

Robot Controllers: Types & Specifications

1. Types of Robot Controllers

- **Teach Pendant Controllers**:
  - Handheld devices for programming and controlling the robot in real-time.
- **Microcontroller-Based Controllers**:
  - Compact, low-cost controllers for basic tasks.
- **Industrial PLC Controllers**:
  - Programmable Logic Controllers for integrating robots with industrial automation.
- **PC-Based Controllers**:
  - Offer high computational power and flexibility.
- **AI-Based Controllers**:
  - Enable adaptive and intelligent control using machine learning and AI.

2. Specifications

- **Processing Speed**: Determines how fast the controller processes instructions.
- **Connectivity**: Interfaces like Ethernet, USB, or fieldbus for integration.
- **I/O Ports**: Number of input/output ports for sensors and actuators.
- **Power Handling**: Voltage and current capacity for driving actuators.
- **Safety Features**: Emergency stops, safe zones, and collision detection.